

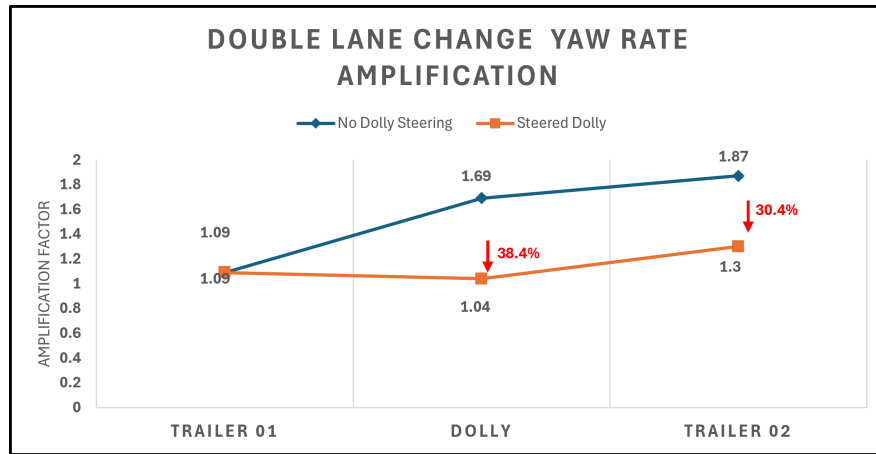


Figure 4.25: Reduction in rearward amplification fo double lane change with dolly steering.

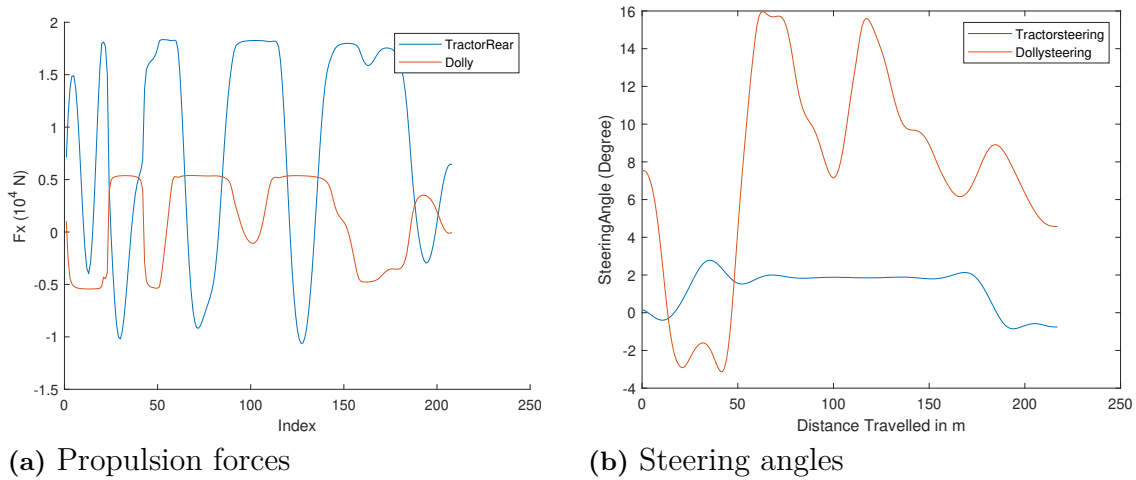


Figure 4.26: Rearward amplification in double lane change when dolly is steered optimally

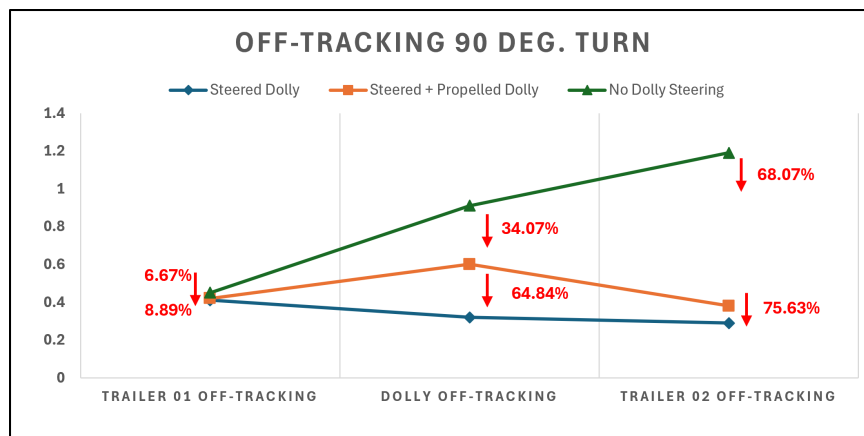


Figure 4.27: Performance comparison for reduction in off-tracking using a propelled dolly.

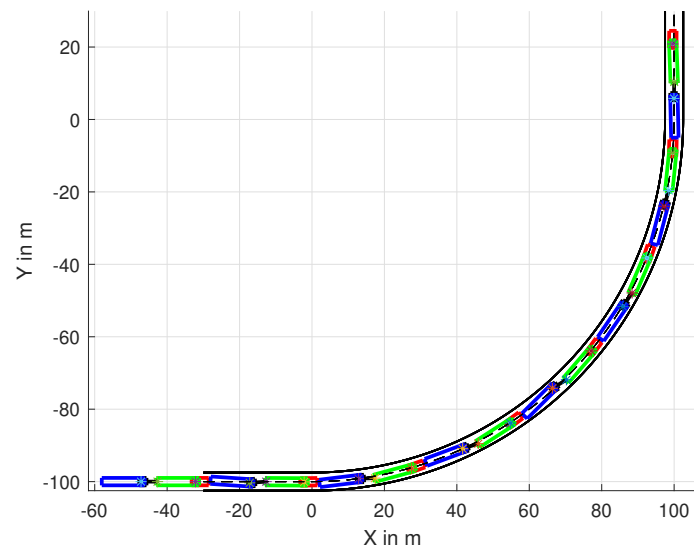


Figure 4.28: Vehicle trajectory for LCV with steered and propelled dolly axle.

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Conclusion

This thesis work proposes an optimal steering control strategy for the tractor and dolly of an A-double combination. This strategy significantly enhances stability, maneuverability, and safety across various driving scenarios. The optimization methods and cost functions utilized in this study can also be adapted for use in other long vehicle combinations with steerable axles.

Two types of vehicle models were employed in this work: kinematic and kinetic, each offering distinct advantages for optimization purposes. The kinematic model is highly effective for finding optimal solution in less time for low-speed maneuvers. Its simplicity allows for faster computations, which is especially beneficial when the number of units in the vehicle combination increases. This model simplifies the control process without compromising the ability to plan and execute precise maneuvers. The kinetic model becomes essential when considering scenarios where vehicle dynamics significantly impact stability and maneuverability. This model incorporates the forces and moments acting on the vehicle, providing a more accurate representation of its behavior at higher speeds or during complex maneuvers. As shown in the results section, both models produced nearly similar outcomes, so the choice between them depends on the maneuver characteristics and computational cost.

The use of a single cost function allows the optimization process by addressing multiple objectives simultaneously. Since these objectives (off-tracking, yaw-amplifications) are interconnected, a unified cost function can efficiently balance them without the need for separate optimization criteria. Overall, the cost function axle mid-point performed well compared to others in terms of achieving multiple objectives in various maneuvers. As it gives freedom to every steerable multiple units The optimal control strategies can assist human drivers in maintaining control and stability, especially in challenging situations. And for autonomous vehicles too, ensuring the objectives are met and vehicle follows the desired path. The process could be extended to reverse parking of LCVs, where precise control and maneuverability are crucial. In vehicles with multiple steerable axles, there is a risk of overactuation, where too many control inputs can lead to inefficiencies or conflicts. The proposed optimization method effectively manages this by coordinating the actions of all steerable axles, ensuring smooth and efficient control. As optimization was performed considering the path distance domain rather than time-controlled actions. This approach is advantageous in scenarios where the vehicle may come to a halt, as it eliminates dependency on time-based controls.

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Discussion

In this section we try to delve into the possible solutions and scenarios on how these cost functions can be implemented in a real time scenarios and be able to perform optimal control allocation.

6.1 Real-Time Implementation

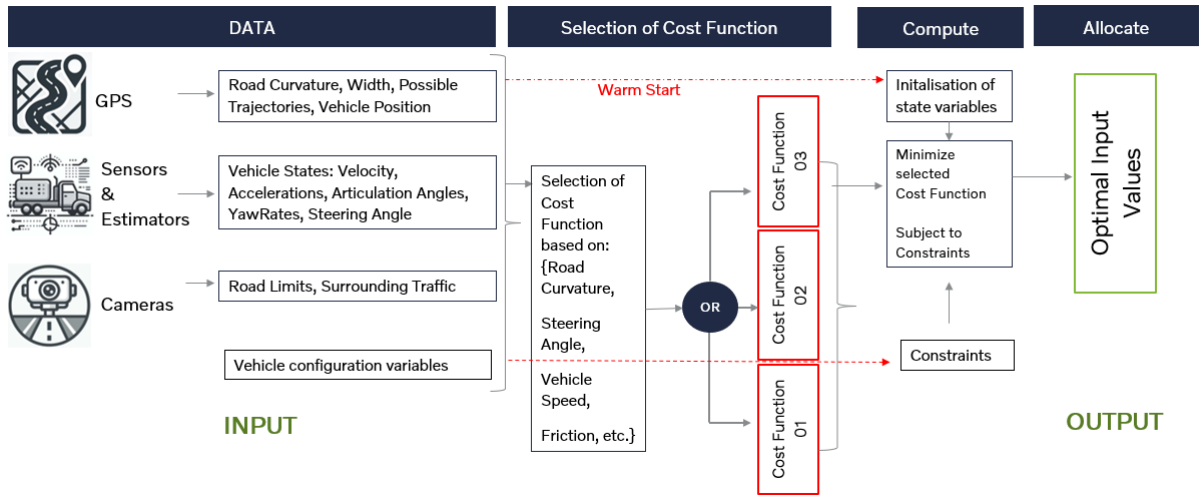


Figure 6.1: Flowchart for real-time implementation.

As seen in the above figure, we can look at the implementation of the cost function in real time by addressing the four different phases.

1. **Input Data:** The input data for the cost function is the curvature of the road ahead, estimated and measured vehicle states, vehicle configuration variables and data about the surrounding environment such as available path and road limits.
2. **Selection of Cost Function:** Once we have the data on road curvature and possible maneuverability options, the system will be able to choose the most suitable cost function for the scenario.
3. **Computation:** After the selection of cost function is done, the optimisation function is evaluated online to compute the optimal control inputs for the different actuators for the upcoming scenario.

4. Allocation: Now, that the system has the computed values for the desired maneuverability. The control inputs are allocated once the vehicle unit reaches, the particular point in the path for which the optimal input was computed.

6.1.1 Collection and usage of Input Data

The input data for the control algorithm can be collected from multiple types of sensors and estimators. For the data of road curvature we use the GPS and satellite map data and route map being used. From this road curvature data, the system will be able to compute the cost function for the section of the path ahead of time and keep the optimal control inputs ready for allocation once the vehicle unit reaches that desired point in the path.

From the path data, the system will also be able to initialise data for estimated vehicle states like yaw angles, position and speed. This data can be supplied to the optimisation algorithm as a warm start to improve computation speed. In addition to this the sensors and estimators onboard the vehicle are also used for information of current vehicle state and other vehicle configuration variables like mass, cog position etc. which is used to determine the constraints like limiting lateral acceleration to avoid unsafe modes.

In addition to the above, camera data is also used to limit actuation if necessary based on surrounding environment conditions.

6.1.2 Selection of cost function

From the initial evaluation of cost functions for different maneuvers, it is evident that a certain cost function can be more suitable for a particular maneuver, so real time computation and comparison of this cost functions online will make it possible to implement the best performing cost function or based on pre-defined maneuver conditions cost function selection can be carried out online.

6.1.3 Implementation of Optimal Input

Once the optimally allocated control inputs are computed, the input signals can be sent to the actuators once the unit combination reaches a desired location on the path for which the control input was computed.

6.2 Future Scope

The strategy to optimally control multi-trailer vehicles can be significantly expanded by increasing the number of actuators, not only for steering but also for propulsion. Further investigating more cost functions and employing improved vehicle model, consequently opens up numerous possibility for further research. The following points outline potential areas of investigation:

1. Advanced vehicle model: The vehicle model used in this thesis can be upgraded to a multi-track model with a non-linear tire model. This improvement will enable a more accurate representation of vehicle dynamics, closely resembling the behavior of actual vehicles.
2. Torque vectoring: Utilizing a multi-track model provides the opportunity to investigate different propulsive forces on either side of the vehicle. This can help in active steering by distributing torque optimally across the wheels.
3. Propulsion on trailer: Researching the effects of adding propulsion systems to trailers can bring insights into better vehicle handling and energy management.
4. Implementation in VTM: The optimal actuation strategy developed can be implemented and tested in high fidelity model (VTM) to evaluate its real-world effectiveness.

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Appendix 1

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